# Similarity-transformed Hamiltonian and ECC diagrams

Signatures are (created particles, created holes, annihilated particles, annihilated holes), all relative to the reference. Open operator legs are short stubs: creation upward, annihilation downward. All arrows are shown in the operator/ECC layout. $T$ uses solid lines and filled circles; $S$ uses wavy lines and filled squares. Crossings without a marker are not contractions.

## Bare Hamiltonian: all thirteen vertices

Vertex 0

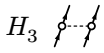
Signature: (1, 0, 1, 0)

Vertex 1

Signature: (0, 1, 0, 1)

Vertex 2

Signature: (0, 0, 1, 1)

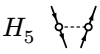



Vertex 3

Signature: (2, 0, 2, 0)

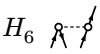
Vertex 4

Signature: (0, 2, 0, 2)



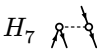

Vertex 5

Signature: (1, 1, 1, 1)




Vertex 6

Signature: (1, 0, 2, 1)




Vertex 7

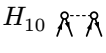
Signature: (0, 1, 1, 2)

Vertex 8

Signature: (2, 1, 1, 0)

Vertex 9

Signature: (1, 2, 0, 1)

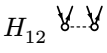



Vertex 10

Signature: (0, 0, 2, 2)

Vertex 11

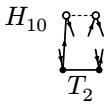
Signature: (1, 1, 0, 0)




Vertex 12

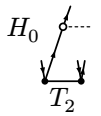
Signature: (2, 2, 0, 0)

## Partial contractions and scalar terms



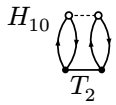

Annihilation sector

Signature: (1, 2, 1, 2)




Unused creation legs

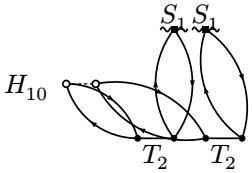
Signature: (2, 2, 0, 0)




Correlation energy

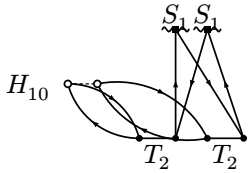
Signature: (0, 0, 0, 0)

## ECC: incidence matters



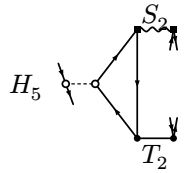

Two single-source $S_1$ vertices

Signature: (0, 0, 0, 0)




Each $S_1$ touches two $T_2$ vertices

Signature: (0, 0, 0, 0)

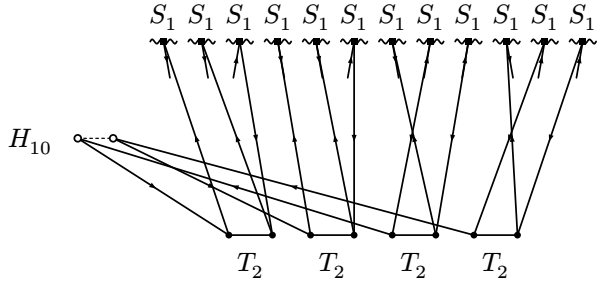



All three edge families; open $S$ legs

Signature: (1, 2, 1, 2)

Both scalar graphs are valid in original ECC coordinates. Only the middle graph meets the stronger doubly-linked energy rule. The renderer does not filter diagrams or change amplitude coordinates.

## The S family is not restricted to four vertices




Four $T_2$ and twelve $S_1$ vertices

Signature: (0, 0, 6, 6)